Online Sports Team Management System SportTrackOTM: A Role-Based Web Platform for Player, Training, Attendance and Performance Records

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ABSTRACT

Many sports teams still run on a mix of paper registers, spreadsheets and group chats. Player details sit in one place, attendance in another and match schedules in a third, so coaches spend a lot of time chasing information instead of coaching. This paper describes a simple web-based system that brings player, team, match, training, attendance and performance records into one database and gives each user a view that matches their role: administrator, coach or player. Administrators manage users and teams, coaches schedule sessions and record attendance and match statistics, and players log in to see their schedules, personal statistics and updates. Attendance is worked out with a plain percentage formula, and dashboards turn the stored records into readable summaries. We keep the design small on purpose. It is a team-level tool for daily management, not a complex tournament engine, and it needs only standard web techniques, not advanced artificial intelligence. The paper explains the architecture, the data model, the main modules and a test plan, and it is open about the limits of the prototype. The system is a learning project and also coordination aid, not a finished commercial product.

KEYWORDS: sports team management; web application; role-based access control; attendance tracking; performance records; relational database; dashboard

1. INTRODUCTION

Running a sports team takes much more than what happens on the field. Someone has to keep the player list current, announce training, note who turned up, record match results and remember how each player has been doing. In many clubs and college teams this is done with paper registers, spreadsheets and messaging apps. Each tool does one job well enough, but none of them talk to each other, so the same information gets copied, goes out of date or gets lost.

The cost is easy to see. A coach who wants to know how regularly a player attends practice has to count marks across several sheets. A player who missed a message about a rescheduled match has no single place to check. An administrator who needs an end-of-season summary has to rebuild it by hand. Each of these is a small annoyance, but together they slow down coordination and make it hard to review how players and the team are really doing.

Web technology can fix most of this without anything exotic. A shared database, a login system and a few wellchosen pages are enough to keep everyone looking at the same facts. Still, a useful system has to answer three questions: who is allowed to see and change what, how the daily numbers such as attendance are calculated, and how easy the system is for a busy coach or a student player to use. These three questions shaped almost every choice we made.

The aim of this work is to design a complete, small team-management platform that covers the daily routine of a sports team. The main contributions are:

- **Centralized records:** one database for players, teams, matches, training sessions, attendance and performance.
- **Role-based access:** Separate views for administrators, coaches and players on their role.

- **Automatic attendance:** Very simple percentage calculation that totally removes manual counting.
- **Readable summaries:** simple dashboards and reports instead of raw tables.
- **Honest scope:** a clear test plan and a plain statement which clarify what the prototype does not do.

3. PROPOSED SYSTEM

3.1 System Architecture

The system is a straight three-layer pipeline with a few side branches. A user signs in, and the web application checks the role and shows only the pages that role is allowed to use. Every action reads from or writes to a single database. Attendance percentages, summaries and reports are produced from those stored records, and schedule changes are passed on to the affected players as notifications. Figure 1 shows how the parts fit together.

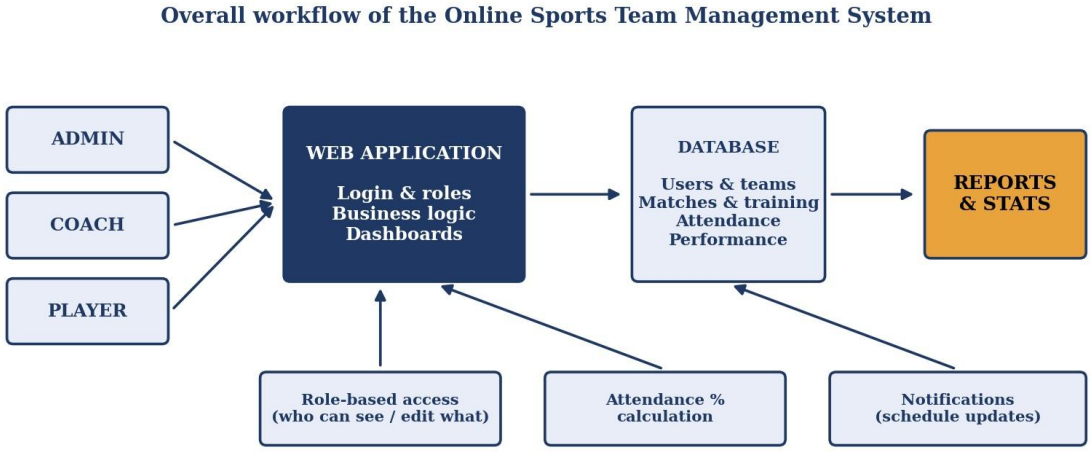


Figure 1: Overall workflow of the proposed system, from users to reports, with the access control, attendance and notification branches.

3.2 User Roles

Three roles are supported. Administrators can manage users, teams and overall settings. Coaches schedule training sessions and matches, mark attendance and enter performance statistics for their own teams. Players can view their own schedule, attendance and statistics, along with team updates. Table 2 lists the permissions of each role.

Table 2: Role-based permissions in the proposed system.

Function	Admin	Coach	Player
Manage users and teams	Full	–	–
Schedule matches and training	Full	Own teams	View
Mark attendance	Full	Own teams	View own
Enter match statistics	Full	Own teams	View own
Post updates / notifications	Full	Own teams	Receive
View reports and dashboards	All	Own teams	Own summary

3.3 Database Design

All information is kept in one relational database so the records stick consistent across all modules. Figure 2 shows the logical data model. Users belong to teams, teams hold training sessions and play matches, attendance records link a player to a training session, and performance records link a player to a match. Because of these links, both individual and team-level summaries can be produced from the same tables without storing anything twice.

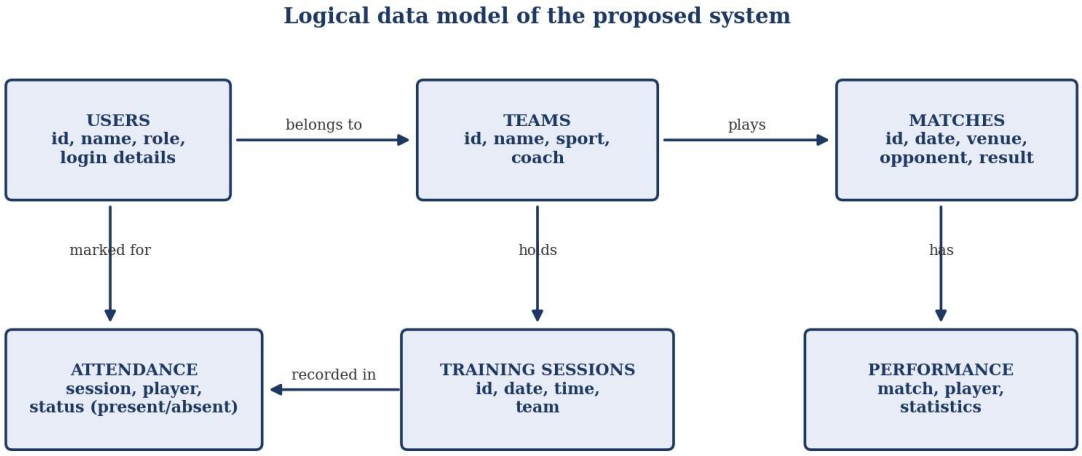


Figure 2: Logical data model of the proposed system (entity names and fields are illustrative and will be matched to the final implementation).

3.4 Software and Hardware Environment

Table 3 lists the tools used to build and run the prototype.

Table 3: Tools and technologies used in the prototype.

Component	Tool / technology and purpose
Front end	HTML, CSS, Bootstrap and JavaScript
Back end	Python with Django Framework
Database	MySQL
Development tools	VS Code , Python virtual environment (venv), pip and MySQL Workbench
Hardware / hosting	Personal Computer, The Django development server (127.0.0.1:8000)

3.5 End-to-End Workflow

A typical week looks like this. The administrator creates the team and adds players and a coach. The coach schedules training and a match, and players see them on their pages at once. After each session the coach marks who was present, and after each match the coach enters statistics. The system then recalculates attendance and refreshes the dashboards, so the next time a player or coach opens the page, the numbers are already up to date.

4. METHODOLOGY

4.1 Authentication and Role-Based Access

Each user has an account with one role. After login, every page and every action is checked against that role, following the permissions in Table 2 and the idea of role-based access control [6]. A coach can change records only for the teams they coach, and a player can only view their own data. Checking the role on the server for every request, not just hiding buttons in the interface, is what actually keeps the data safe.

4.2 Record Management and Filtering

All records are handled with the four basic operations: create, read, update and delete (CRUD). Filtering lets a user find records quickly, for example all sessions of one team in a given month, or all matches against one opponent. Deleting a player or a team is restricted to the administrator, and linked records are handled so that attendance and performance history are not left pointing at nothing.

4.3 Scheduling and Notifications

Coaches and administrators create matches and training sessions with a date, time and venue. When a schedule is created or changed, the update becomes visible to the players of that team and can be shown as a notification. Our prototype does not decide match pairings itself. Fair fixture generation, as in RoundGen [1], is a separate problem that we list under future scope.

4.4 Attendance Calculation

Attendance is the number reviewed most often in daily team life, so it is calculated automatically. For one player, the attendance percentage is the number of sessions attended divided by the total number of sessions held, multiplied by 100:

$$Attendance\ (\%) = (P / N) \times 100 \tag{1}$$

Here P is the number of sessions in which the player was marked present and N is the total number of sessions held. For example, a player who was present at 18 of 20 sessions has an attendance of 90%. Algorithm 1 gives the steps, including the case where no sessions have been held yet.

Algorithm 1: Attendance Percentage Calculation

```
Input: Total sessions N, present sessions P for a player
Output: Attendance percentage A
Read N from the session records
Count P from the attendance records
if N = 0 then
    A ← 0
else
    A ← (P / N) × 100
end if
Display A on the player and coach dashboards
return A
```

Algorithm 1: Attendance percentage calculation.

The same formula, applied to all players of a team together, gives the team attendance. A second simple figure, the win rate of a team, is the number of matches won divided by the number of matches played, multiplied by 100:

$$Win\ rate\ (\%) = (W / M) \times 100 \tag{2}$$

4.5 Performance Records

Match-level statistics are saved per player and per match. Which statistics are stored depends on the sport, for example points, goals, assists or minutes played. Keeping them at match level means they can later be added up into season totals and averages without losing detail. Players can see their own figures, and coaches can compare players within their team.

4.6 Dashboards and Reports

Instead of showing raw tables, the system gives each role a short summary. Coaches see team attendance and recent match results. Players see their own attendance percentage, upcoming sessions and statistics. Administrators is only who overview of teams and users. The aim is that nobody has to open a spreadsheet to answer a simple question.

5. EXPERIMENTAL SETUP AND EVALUATION FRAMEWORK

The prototype is tested with sample data: a few teams, a set of players, a coach and an administrator account, and a series of training sessions and matches. This section explains how the system will be checked and how the results should be read.

5.1 Expected Failure Modes

Before running the tests we wrote down the problems we expected. Wrong or missing attendance entries will give wrong percentages, however correct the formula is. Two coaches editing the same record at the same time may overwrite each other. A role check that is done only in the interface, and not on the server, could let a user reach a page by typing its address. Players with similar names may be mixed up if records are searched by name instead of by a unique identifier. Each of these is covered by at least one test in Table 4.

6. RESULTS AND DISCUSSION

6.1 Test Results

This paper reports the design of the prototype and its evaluation plan rather than field results, and we do not present any measured figures that have not been collected. The system will be evaluated against a structured test plan, summarized in Table 3, using sample teams, players and sessions.

Table 5: Results of the prototype tests (to be filled after testing).

Test	What is checked	Outcome
CRUD operations	Records can be created, viewed, edited and deleted for users, teams, matches and sessions	Accurate records after every operation
Attendance calculation	Computed percentage matches manual calculation for sample data, including zero session case	Correct values, no errors
Access control	Each role can reach only its permitted pages and actions	Unauthorized actions blocked
Scheduling visibility	Newly created or changed sessions and matches appear for the right users	Updates visible to players
Performance records	Match statistics are stored and retrieved per player	Consistent, retrievable records

6.2 Discussion

What we expect, and what the tests let us check, is that the system keeps records accurate, that each role sees only what it should, and that coordination needs less manual effort than the paper and chat approach. The attendance calculation is simple enough to verify by hand, which makes it a good first test. Access control needs the most care, because an error there can expose or damage other people's data. We also note that a system like this only helps if people actually use it. A coach who keeps marking attendance on paper will make the dashboard useless, so ease of use matters as much as correctness.

6.3 Advantages

- Team information is in one place instead of instead files and chats.
- Manual record keeping is reduced, especially for attendance.
- Schedules and updates are easy for players to find.
- Attendance and performance data are organized and easy to review.
- Role-based control protects sensitive data and operations.
- The design can be extended to other sports.

6.4 Limitations

- The system depends on accurate data entered by users. Wrong input can make gives wrong summaries.
- Deployment needs the network access.
- Advanced live scoring is outside the scope of the basic prototype.
- Notification features depend on how they are implemented.

- Because personal data about players is stored, security needs careful attention before real use, including secure password storage and protection of the database.
- The prototype has been tested only with sample data, not with a real team over a full season.

7. FUTURE SCOPE

There are several clear next steps. The system can be extended to more sports and team types, with sport-specific statistics. Richer analytics and performance tracking can show trends over a season. Notifications can be extended to e-mail or mobile messages. Live-scoring features can go beyond the basic prototype, and an automatic fixture generator, along the lines of [1], can be added. Finally, the design can be scaled for clubs and institutions with many teams, with stronger security and a trial with a real team.

8. CONCLUSION

This paper presented the Online Sports Team Management System, a small web platform that centralizes the daily management of a sports team. It brings player, team, match, training, attendance and performance records into one role-based system and addresses manual record keeping, scattered communication and weak performance tracking. The prototype uses straightforward web techniques and a simple attendance formula, which keeps it easy to build, understand and maintain. The work also shows how much depends on good data entry and careful access control. The system is meant for learning and as a coordination aid. Any real deployment would need a full test with a real team and proper security review.

Disclaimer: this is an educational research project. The system is a prototype and has not been validated for production use.

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